

Q1 $O^a \oplus O^b : a+b=c$

$$S = O^p 1^p O^{2p}$$

O^α	O^β	$O^{p-\alpha-\beta} 1^p O^{2p}$
n	y	z

Wyz: $O^\alpha O^{i\beta} O^{p-\alpha-\beta} 1^p O^{2p}$
 $= O^{p+i\beta-\beta} 1^p O^{2p}$

now, $p+i\beta-\beta+p \neq 2p$

$\Rightarrow i\beta-\beta \neq 0$

$\Rightarrow \beta(i-1) \neq 0$

$\therefore i \neq 1$

Q2 ww

$$S = O^P \perp O^P \perp$$

$$\begin{array}{c|c|c} O^\alpha & O^\beta & O^{P-\alpha-\beta} \perp O^P \perp \\ \hline \alpha & \beta & \gamma \end{array}$$

~~neg i z~~ $O^\alpha O^\beta O^{P-\alpha-\beta} \perp O^P \perp$

ny i z $= O^{P+i\beta-\beta} \perp O^P \perp$

now, $P+i\beta-\beta \neq P$

$$\Rightarrow i\beta - \beta \neq 0$$

$$\therefore i \neq 1$$

$$\sum 0^n 1^m, n > m$$

$$s = 0^{p+1} 1^p$$

0^{α}	0^{β}	$0^{p+1-\alpha-\beta} 1^p$
x	y	z

$$\frac{xy^i z}{0} = 0^{p+1+i\beta-\beta} 1^p$$

$$\text{now, } p+1+i\beta-\beta \leq p$$

$$\Rightarrow \beta - \beta \leq -1$$

$$\Rightarrow \beta(i-1) \leq -1$$

$$\Rightarrow \beta - i\beta \geq 1$$

$$\text{for } i=0, 1,$$

$$\Rightarrow \beta(1-i) \geq 1$$

$$\therefore i=0, \beta \geq 1$$

4.

$$S = 1^{n^2} + 1^{p^2}$$

1^a	1^b	$1^{p^2 - a - b}$
n	y	z

$$\begin{aligned} xy^iz : & 1^{p^2 + ip - b} \\ (122) &= 1^{p^2 + 2p - b} \\ &= 1^{p^2 + b} \end{aligned}$$

$$\left\{ \begin{array}{l} 1st \rightarrow 1^{p^2} \\ 2nd \rightarrow 1^{(p+1)^2} \end{array} \right.$$

$$\begin{aligned} \text{now, } (p+1)^2 &> p^2 + b \\ \Rightarrow p^2 + 2p + 1 &> p^2 + b \\ \Rightarrow 2p + 1 &> b \end{aligned}$$

$$\begin{aligned} \text{now, } |xy| &\leq b \\ \therefore |a+b| &\leq b \end{aligned}$$

$$\text{so, obviously } b < 2p + 1$$

(proved)

Q5 $\{1^{2^n}, n \geq 0\}$

$s = 1^{2^p}$

1^A	1^B	$1^{2^p - A - B}$
n	2	2

$\text{nyiz: } 1^{2^p + 1^B + B}$

$(A=2) = 1^{2^p + B}$

$\left\{ \begin{array}{l} 1^{st} \rightarrow 1^{2^p} \\ 2^{nd} \rightarrow 1^{2^{p+1}} \end{array} \right.$

now, $2^{p+1} > 2^p + B$

$\Rightarrow 2^p \cdot 2^p > 2^p + B$

$\therefore 2^p > B$

(proved)

Q6 1^p prime

1^q , such that q is

1^A	1^B	1^{q-A-B}
n	2	2

$\text{nyiz: } 1^{q + 1^B + B}$

So, $q + B(i-1)$

So, $i = q+1,$

$q + B(q+1-1)$

$= q + qB = q(1+B)$

now, it can't
 be prime as
 $(q \times (1+B))$, so it
 is not prime.

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ω is not palindromic

$$s = 0^p 1 0^p$$

$$\begin{array}{c|c|c} 0^\alpha & 0^\beta & 0^{p-\alpha-\beta} 1 0^p \\ \hline \alpha & \beta & \ell \end{array}$$

xy²z : $0^{p+i\beta-\beta} 1 0^p$

now, $p+i\beta-\beta \neq p$

$$\Rightarrow i\beta - \beta \neq 0$$

$$\Rightarrow \ell \neq 1$$

$$2. \mathcal{L} = \{ 0^n 1^m \mid n \neq m \}$$

$$S = 0^p 1^{p+p!}$$

0^α	0^β	$0^{p-\alpha-\beta}$	1^p
γ	γ	2	

xyz: $0^\alpha 0^\beta 0^{p-\alpha-\beta} 1^{p+p!}$

$$S_0, p + \beta - \beta = p + p!$$

$$\therefore \bar{\lambda} = \frac{p!}{3} + 1 \quad \left\{ \begin{array}{l} \text{fact always constant} \\ \beta, \text{ this always int} \end{array} \right.$$

not palindrome

L9: $0^n 1^m$

$n = 4q + r_1$ $m = 4q + r_2$

xy^iz

0^α	0^β	$0^{4q+r_1-\alpha-\beta} 1^{4q+r_2}$
x	y	z

So, xy^iz : $0^{4q+r_1+i\beta-\beta} 1^{4q+r_2}$

now, $(i=2)$ $0^{4q+r_1+2\beta-\beta} 1^{4q+r_2}$
 $= 0^{4(q+\beta)+r_1} 1^{4q+r_2}$

So, here $q+\beta \neq q$

hence, it is not proved to be regular.

L10 $0^n 1^m$, $\frac{n}{m}$ is integer

So, $0^{4p} 1^{2p}$

0^α	0^β	$0^{4p-\alpha-\beta} 1^{2p}$
x	y	z

So, xy^iz = $0^{4p+i\beta-\beta} 1^{2p}$

So, $i=2$, $\frac{4p+2}{2p}$, this can't be integer.

CFG

$$L_1 = \{0^n 1^m \mid n = 3m\}$$

$$S \rightarrow A S B B B \mid \epsilon$$

$$A \rightarrow 0$$

$$B \rightarrow 1$$

$$\epsilon \rightarrow \epsilon$$

$$L_2 = \{0^n 1^m \mid n \leq 3m\}$$

$$S \rightarrow A_1 A_2 A_3 \mid \epsilon$$

$$A_1 \rightarrow 0 A_1 111 \mid \epsilon$$

$$A_2 \rightarrow 0 A_2 11 \mid \epsilon$$

$$A_3 \rightarrow 0 A_3 1 \mid \epsilon$$

26: $0^i 1^j 2^k$ $i+k=j$

$S \rightarrow 0S1 \mid A$

$A \rightarrow 1A2 \mid \epsilon$

$$L_3 = 0^i 1^j 2^k \quad i+j=k$$

~~1x~~

∴

$$S \rightarrow 0 S X 2 \mid \epsilon$$

$$X \rightarrow 1 2$$

$$L_4 = 0^i 1^j 2^k ; 2(i+j) = k$$

$$S \rightarrow 0 S A 2 2 \mid \epsilon$$

$$A \rightarrow 1 2 2$$

$$\# L5 = 0^i 1^j 2^k \quad i+j = 2k$$

$$S \rightarrow 00S2 \mid A$$

$$A \rightarrow 11A2 \mid \epsilon$$

01011010

(0111)(1101)

17: every 0 followed by even 1's

$$S \rightarrow A_1 0 A_1 \mid \epsilon$$

$$A \rightarrow 11A \mid \epsilon$$

$$A_1 \rightarrow 1A_1 \mid \epsilon$$

18: w = palindrome where no consecutive symbols

$$S \rightarrow 0S1 \mid \epsilon$$

$$S_1 \rightarrow 1S1 \mid \epsilon$$

19 equal number of 0's and 1's

$$S \rightarrow 1S0 \mid 0S1 \mid SS \mid \epsilon$$

$L10: n \neq y, y \text{ contains exactly } 2 \text{ 1's, } |x|=|y|$

$$S \rightarrow 0S0 \mid 1S0 \mid 0A0 \mid 0A1$$

$$A \rightarrow 0A0 \mid 0A0 \mid 0B1 \mid 1B1$$

$$B \rightarrow 0B0 \mid 1B0 \mid \epsilon$$

Problem 3

'aabbababa'

a)

$S \rightarrow PQ$

$\rightarrow Da**b**Q$

$\rightarrow aabQ$

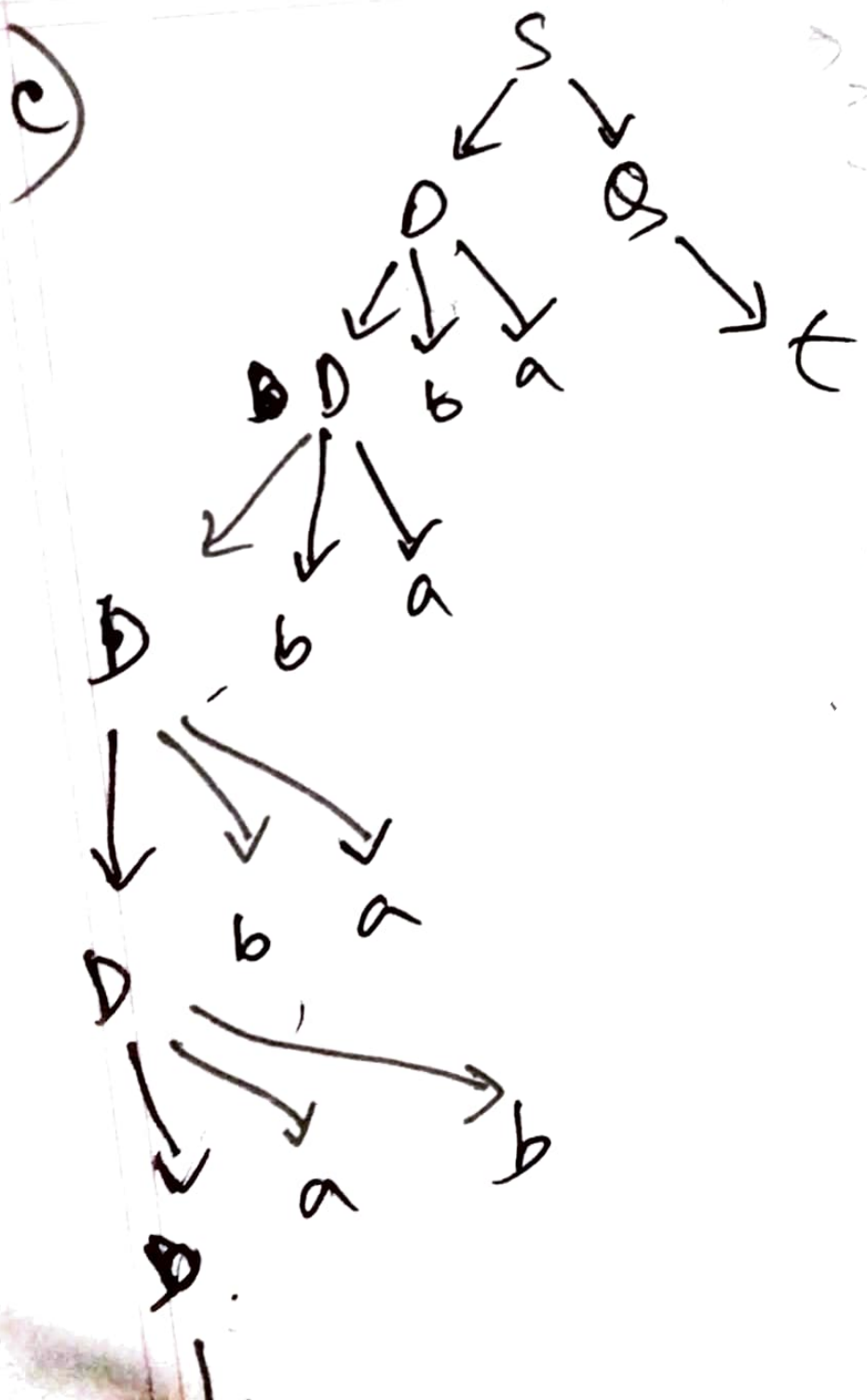
$\rightarrow aab**b**Qa$

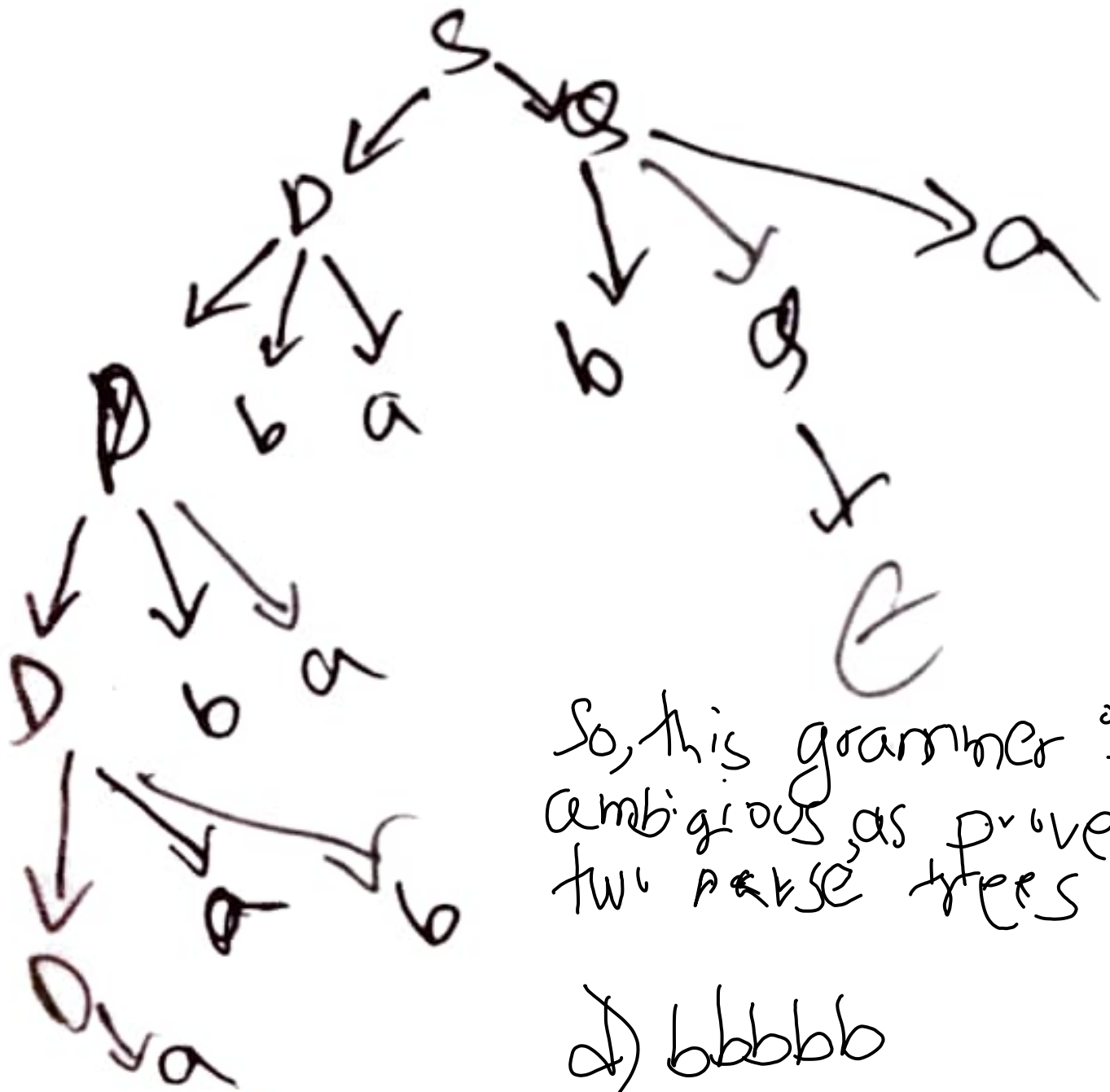
$\rightarrow aabba**b**Qba$

$\rightarrow aabba**b**Qab**a**$

$\rightarrow aabba**b**ab**a**$

b)





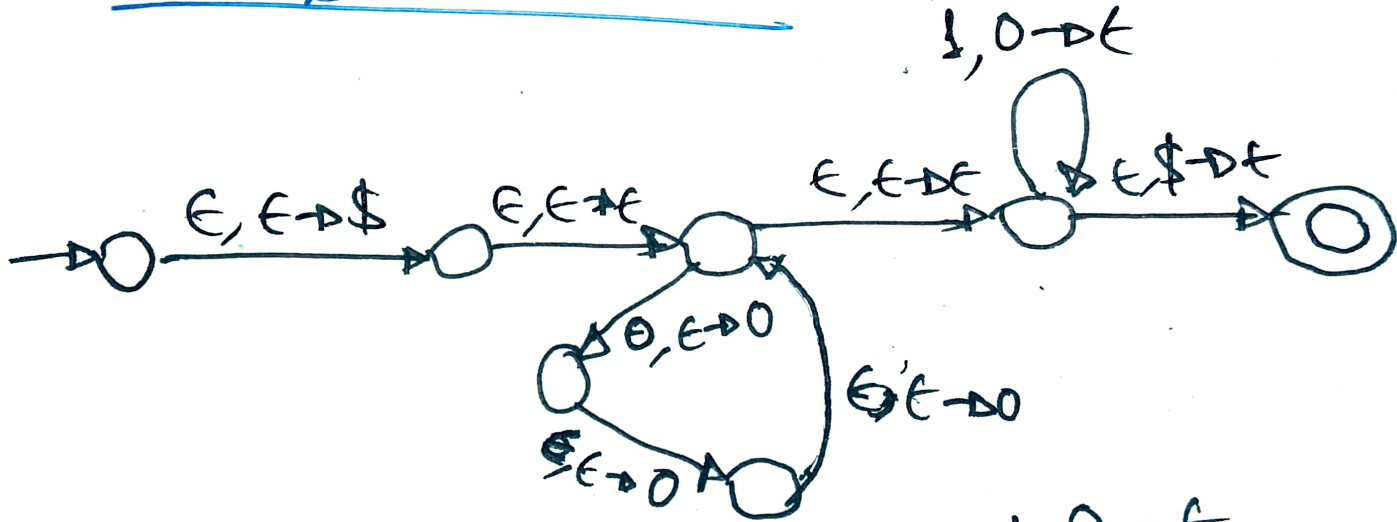
So, this grammar is ambiguous, as proven by two parse trees

d) bbbbbb

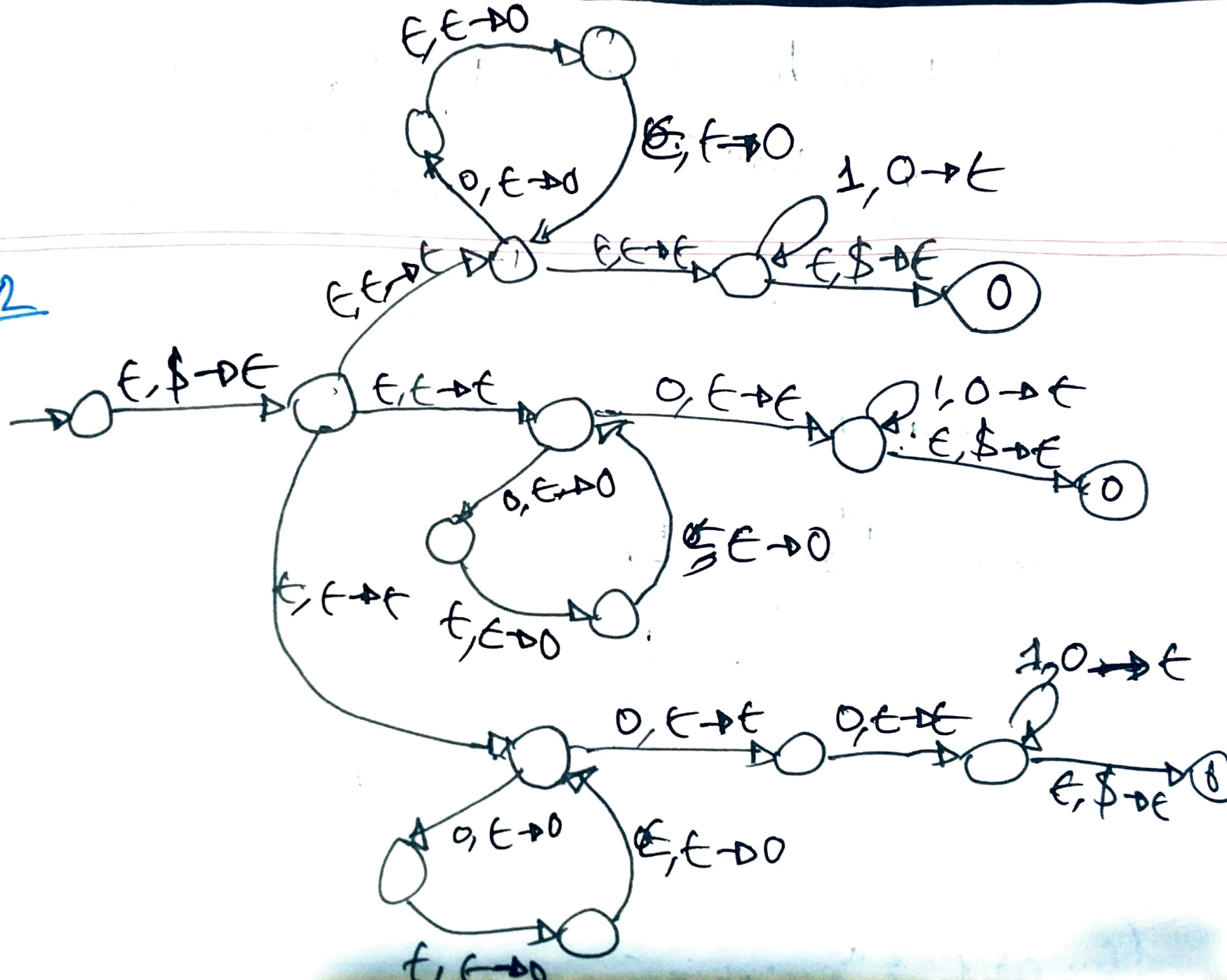
~~1. E-20~~

Ans to 9

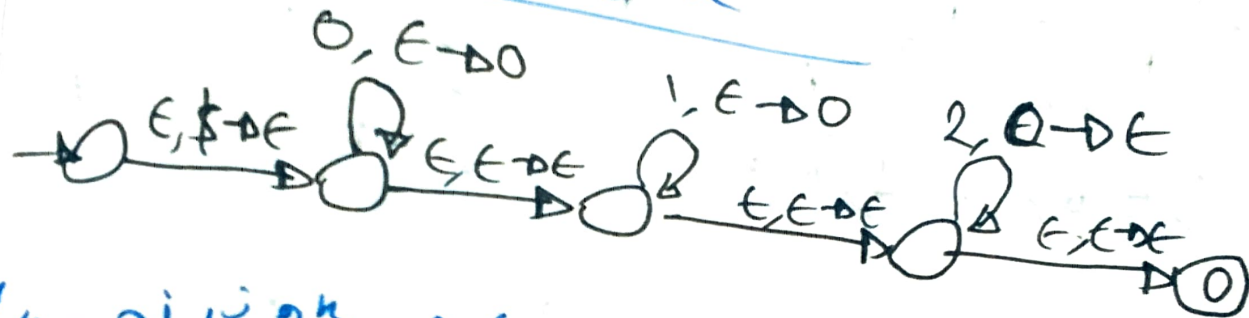
21


$$1, 0 \rightarrow t$$
 ~~$f, e \rightarrow f \rightarrow e$~~

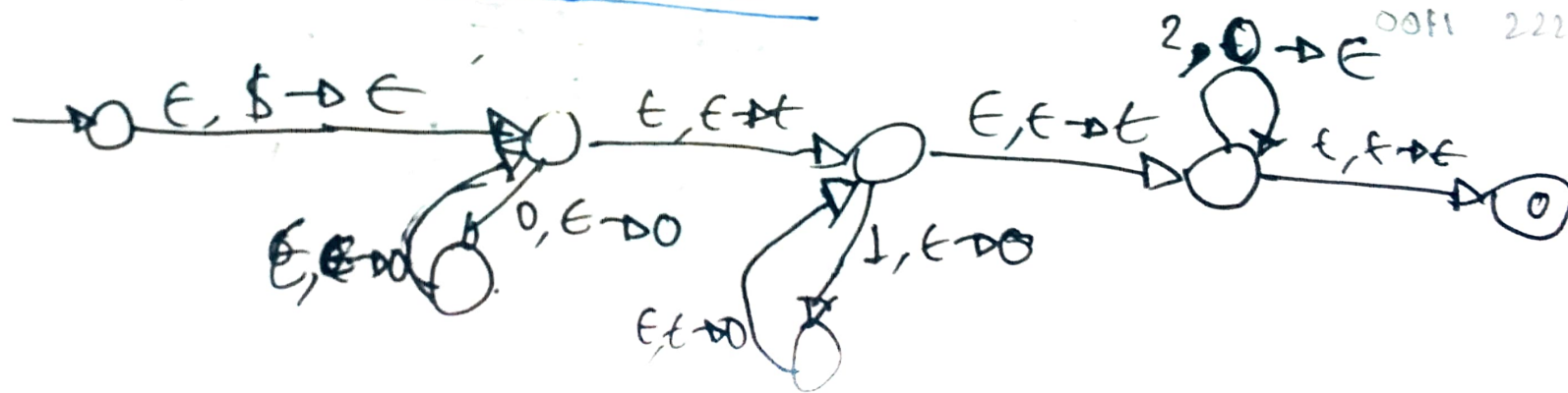
12



1 tick

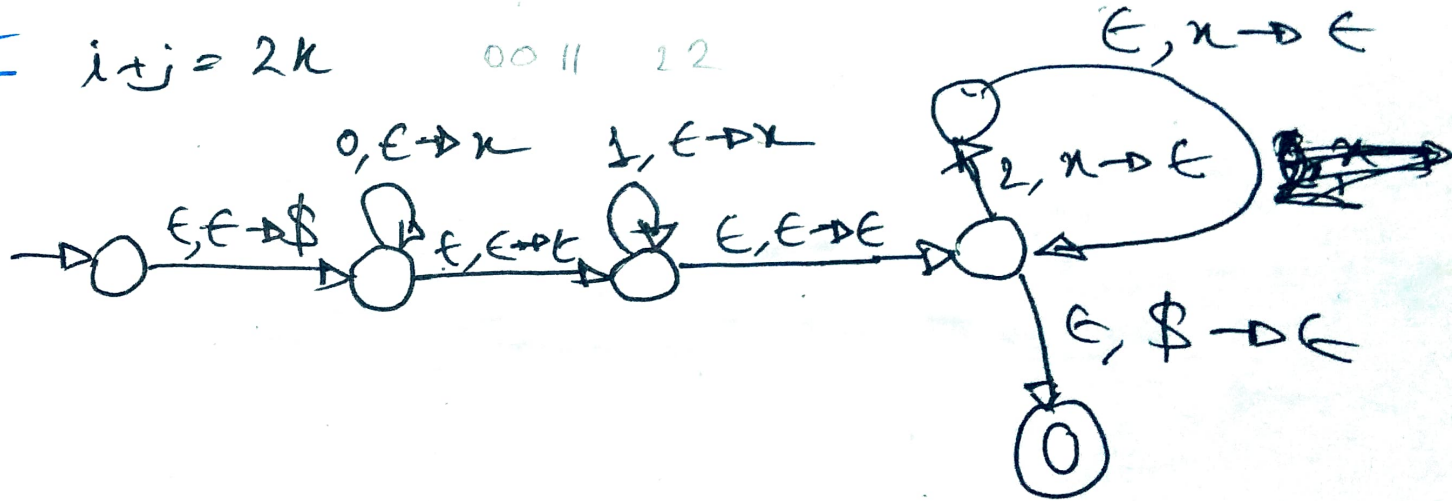


24: $0^i 1^j 2^k$ $2(i+j) = k$

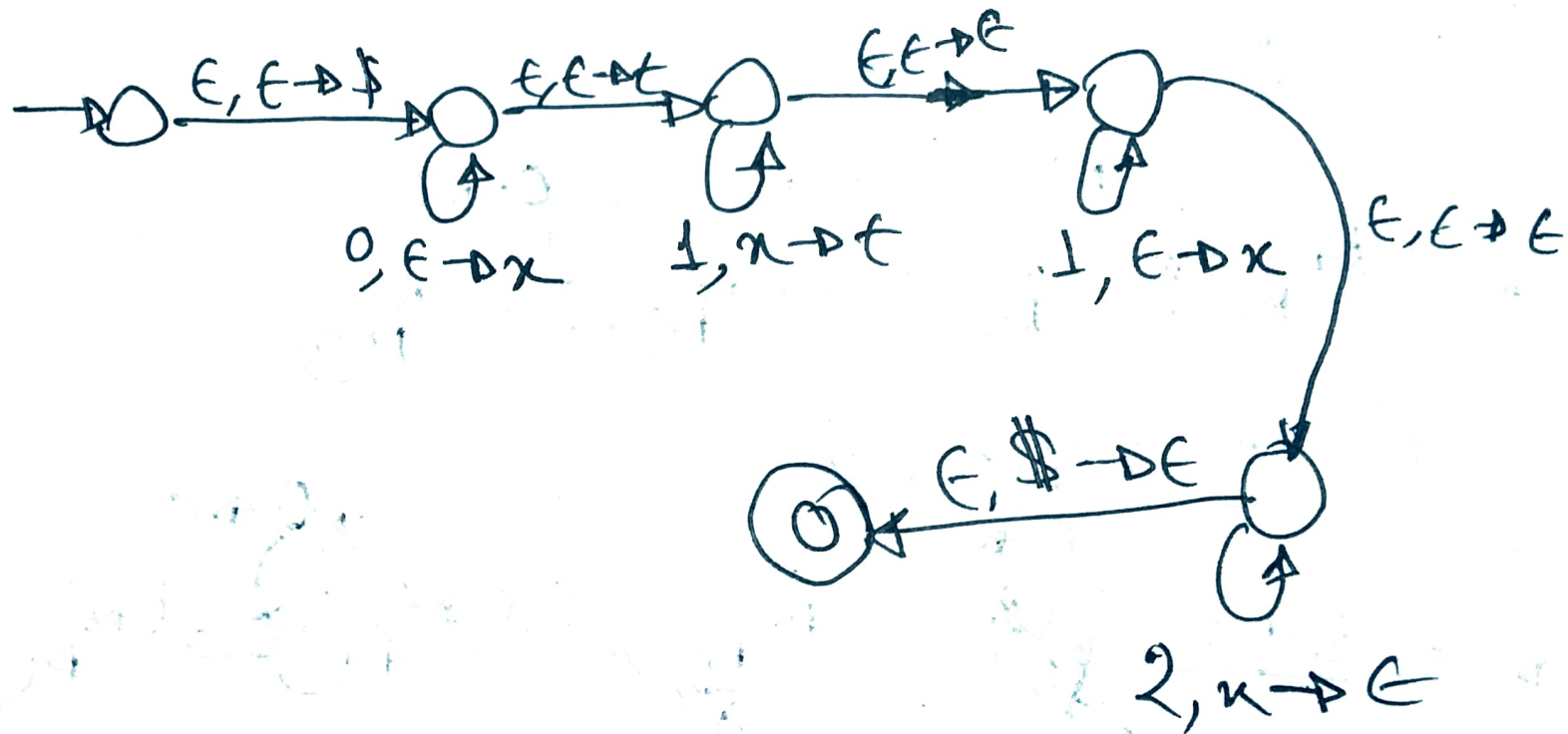


25 $i+j=2k$

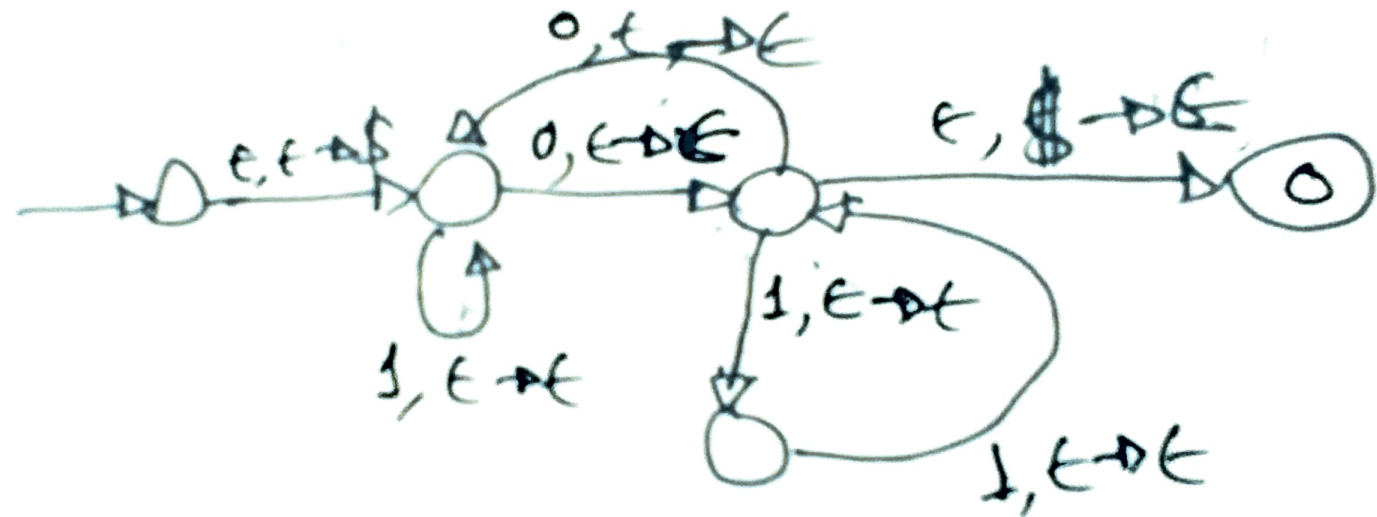
00 11 22



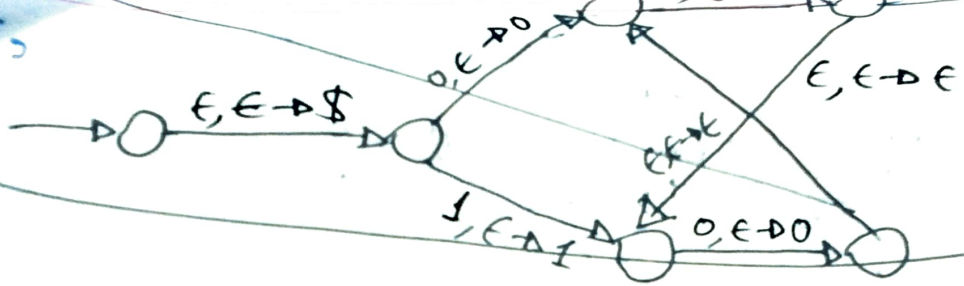
LG: $0^i 1^j 2^k$ $i+k=j$



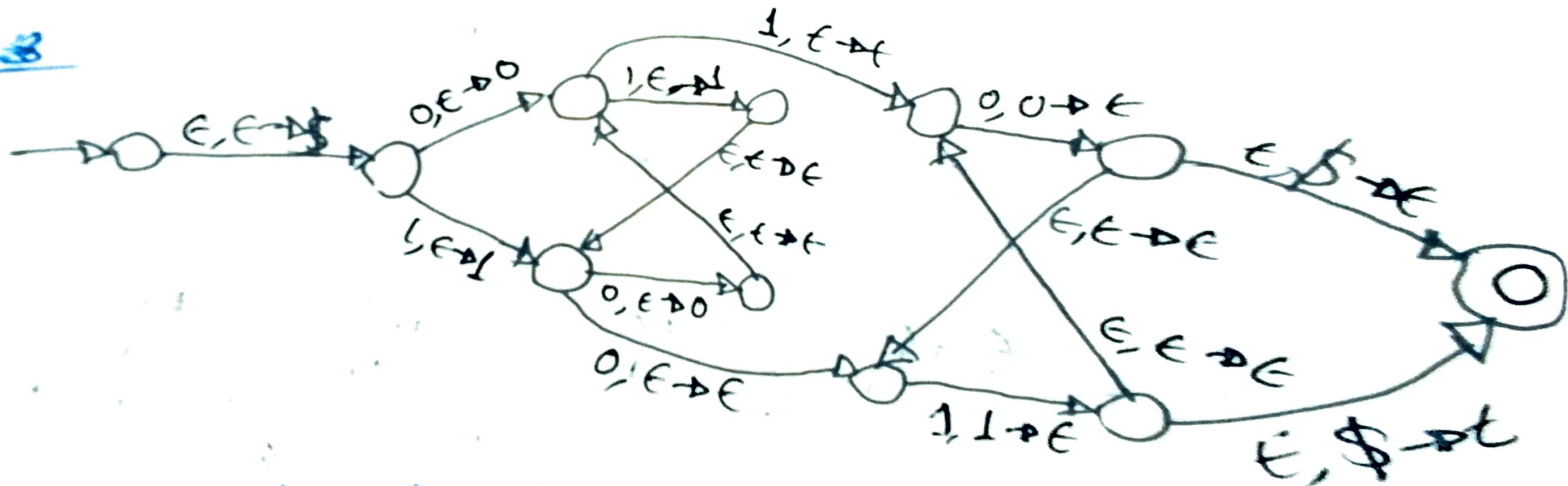
7. 0 in w followed by even 1 's



28,



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equal number of 0's and 1's

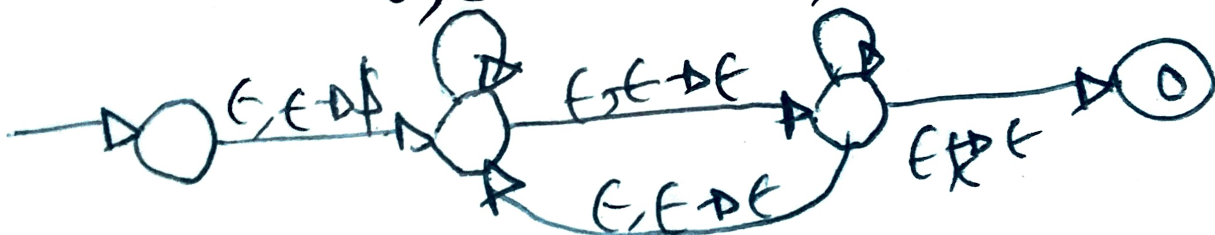
L9 equal 0 and 1

$1, \epsilon \rightarrow 1$

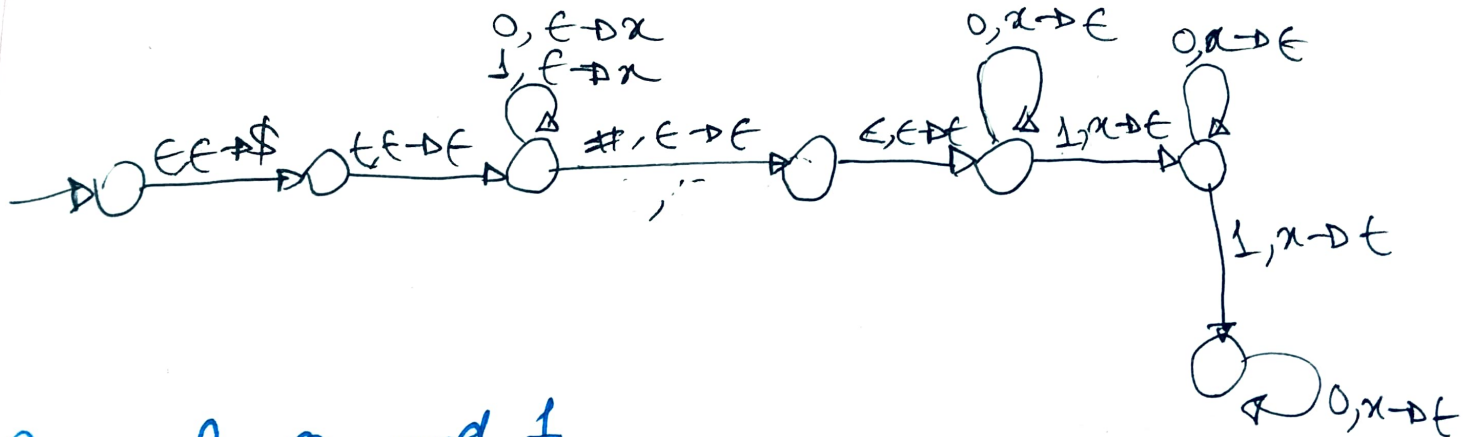
$0, \epsilon \rightarrow 0$

$1, 0 \rightarrow \epsilon$

$0, 1 \rightarrow \epsilon$



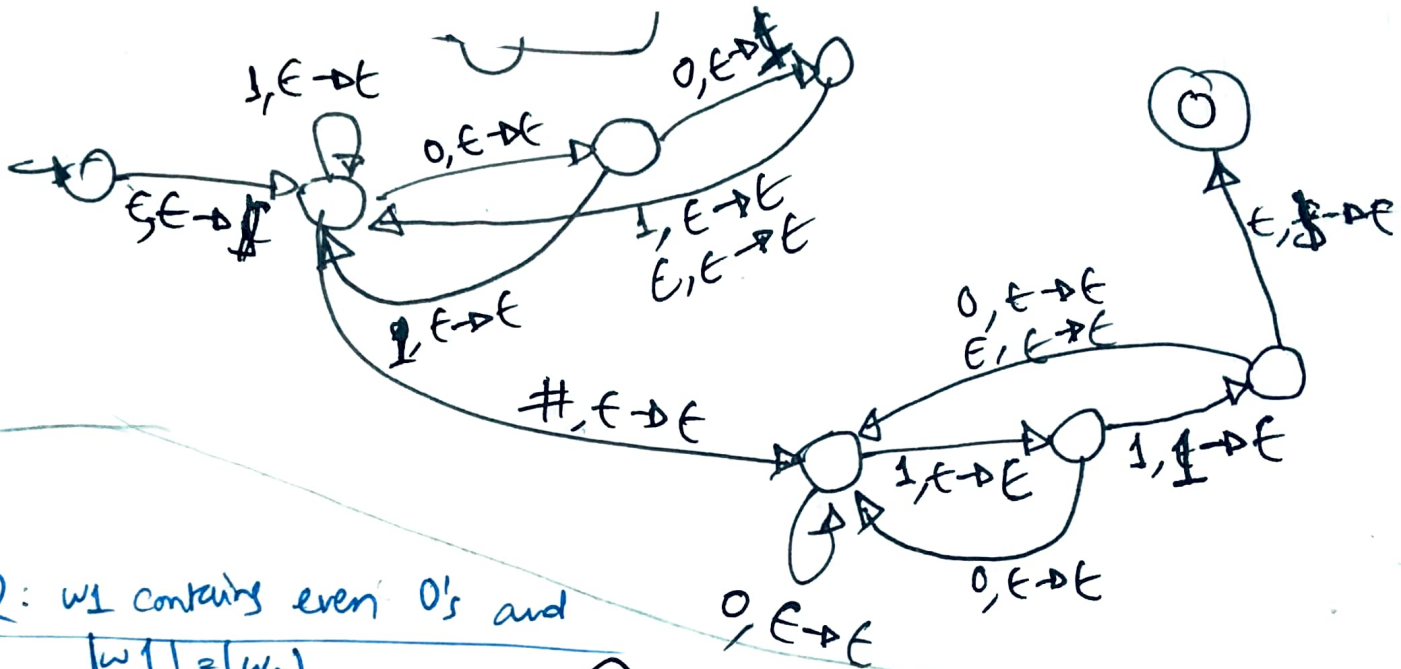
L10 $x \neq y$, y contains two 1s, $|x| = |y|$



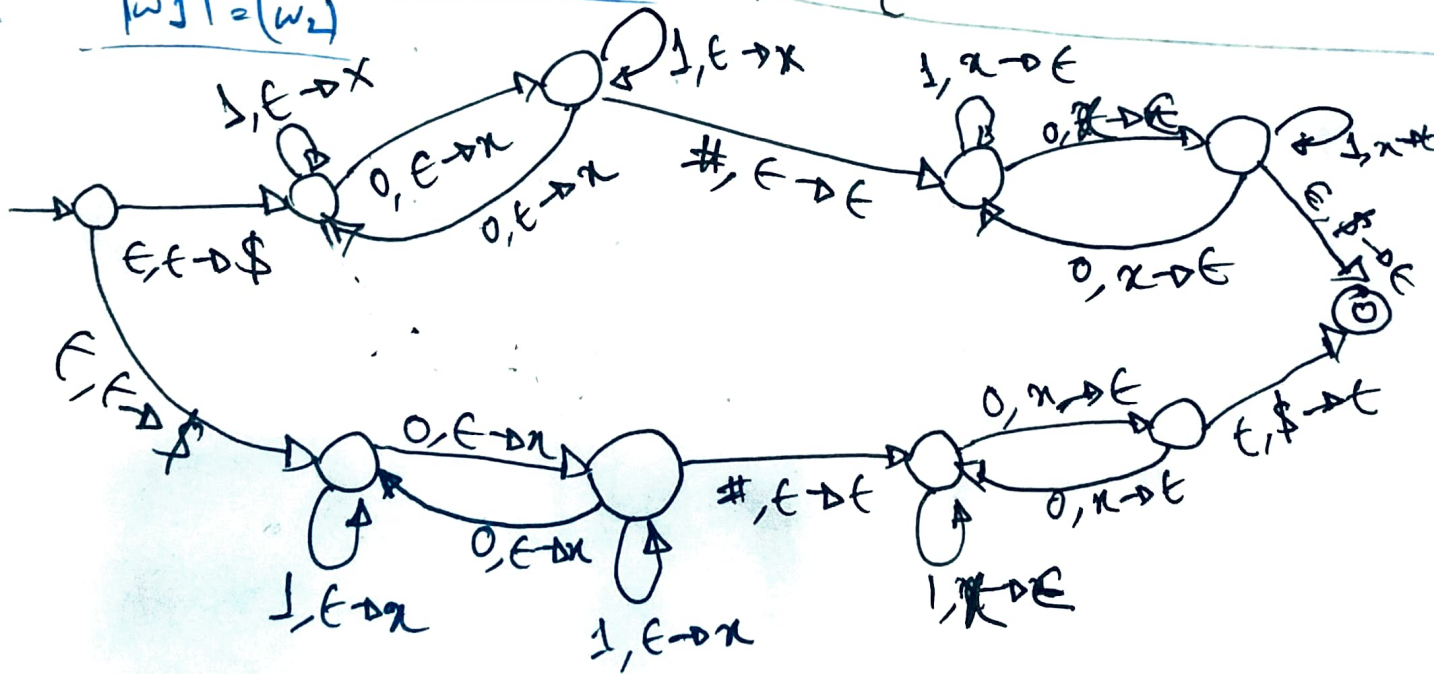
L9 equal 0 and 1

$1, \epsilon \rightarrow 1$ $1, 0 \rightarrow \epsilon$

111 $w_1 \neq w_2$ number of 0's and 1's equal

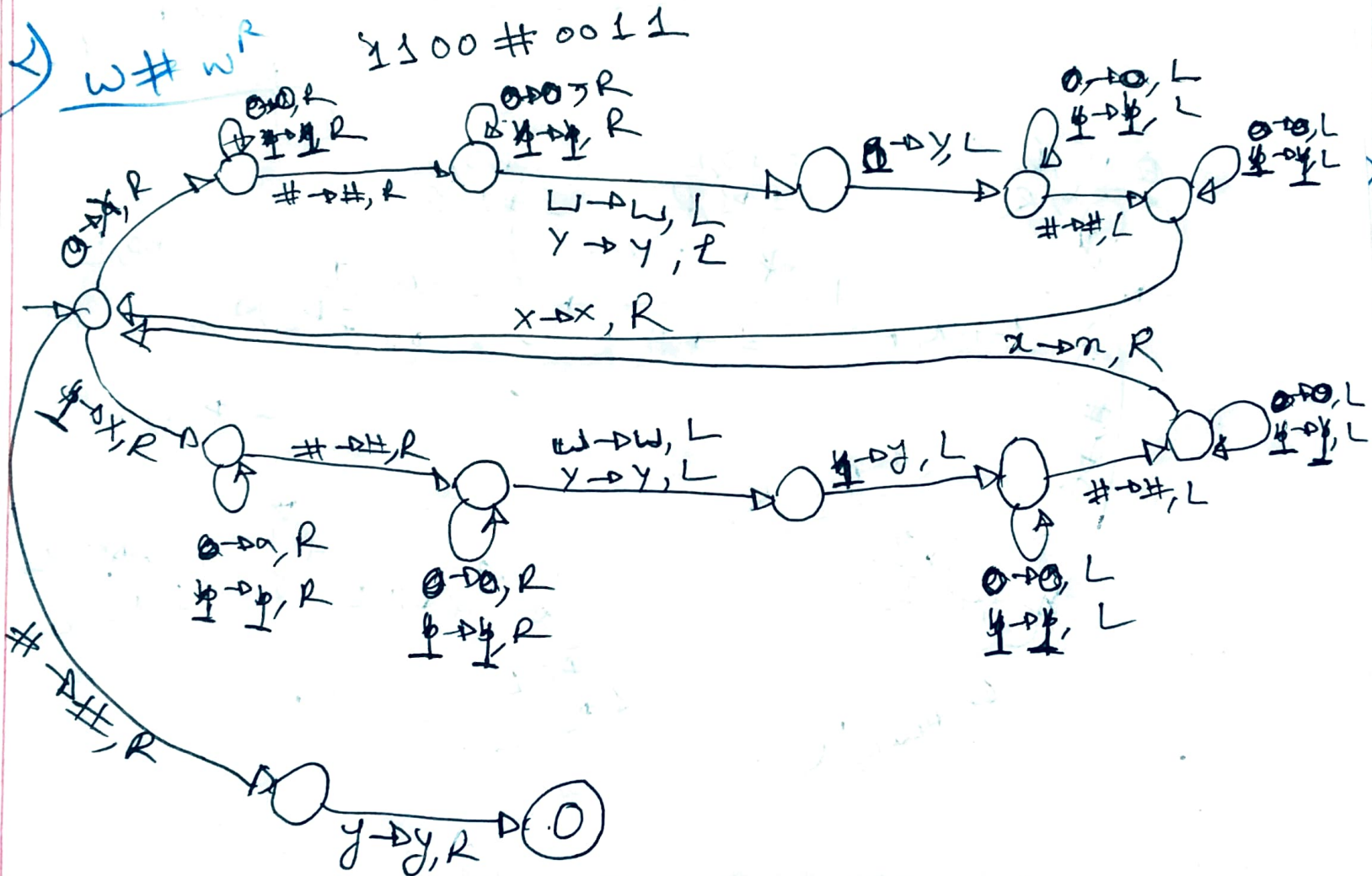
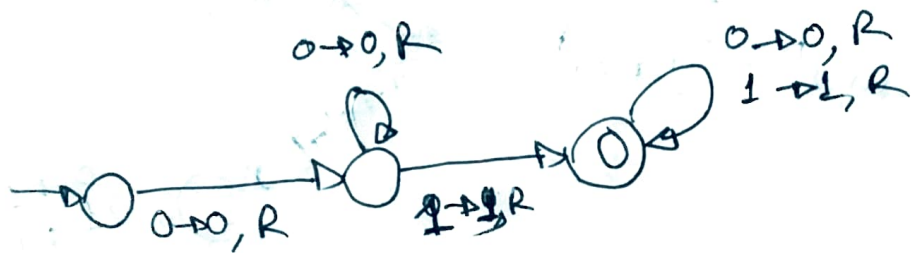


112: w_1 contains even 0's and $|w_1| = |w_2|$



Ans to 5

1) w contains 01



001101 # 111

3) $w_1 \# w_2 \rightarrow \text{count}(w_1, 1) = \text{count}(w_2, 1)$

